

In [51]:

```
import pandas as pd
import matplotlib.pyplot as plt

salaryDataFrame = pd.read_csv('salary.csv')
```

In [52]:

```
fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(salaryDataFrame['YearsExperience'], salaryDataFrame['Salary'])
ax.set_title("Experience vs Salary")
ax.set_xlabel("YearExperience")
ax.set_ylabel("Salary")
```

Out[52]:

Text(0, 0.5, 'Salary')



YearsExperience	Salary
1	40000
1	45000
2	42000
2	44000
3	55000
3	58000
3	60000
3	65000
4	55000
4	56000
4	58000
4	63000
5	58000
5	60000
5	65000
5	68000
6	81000
6	94000
7	91000
7	98000
8	101000
8	114000
9	105000
9	109000
10	112000
10	117000
11	121000
11	123000

In [53]:

```
xValues = salaryDataFrame['YearsExperience']
yValues = salaryDataFrame['Salary']
```

In [54]:

```
xMean = xValues.mean()
yMean = yValues.mean()

print(f"Mean of X: {xMean}")
print(f"Mean of Y: {yMean}")
```

Mean of X: 5.413333333333332
Mean of Y: 76004.0

In [55]:

```
# (xi - xMean) and (yi - yMean)

salaryDataFrame['xDeviation'] = xValues - xMean
salaryDataFrame['yDeviation'] = yValues - yMean
```

In [56]:

```
# (xi - xMean)(yi - yMean)
salaryDataFrame['productOfDeviations'] = salaryDataFrame['xDeviation'] * salaryDataFrame['yDeviation']
```

In [57]:

```
# (xi - xMean)^2
salaryDataFrame['squaredDeviation'] = salaryDataFrame['xDeviation'] ** 2
```

In [58]:

```
numeratorSum = salaryDataFrame['productOfDeviations'].sum()
denominatorSum = salaryDataFrame['squaredDeviation'].sum()
```

In [59]:

```
slopeValue = numeratorSum / denominatorSum
yIntercept = yMean - (slopeValue * xMean)

print("\nCalculation Table (First 5 rows)")
columnsToDisplay = [
    'YearsExperience',
    'Salary',
    'xDeviation',
    'yDeviation',
    'productOfDeviations',
    'squaredDeviation'
]
print(salaryDataFrame[columnsToDisplay].head())
print("\nFinal Regression Parameters ")
print(f" Numerator Sum: {numeratorSum}")
print(f" Denominator Sum: {denominatorSum}")
print("-" * 30)
```

```
print(f"Slope (m): {slopeValue}")
print(f"Intercept (b): {yIntercept}")
print(f"Equation: y = {slopeValue:.2f}x + {yIntercept:.2f}")
```

Calculation Table (First 5 rows)

	YearsExperience	Salary	xDeviation	yDeviation	productOfDeviations	\
0	1.2	39344.0	-4.213333	-36660.0	154460.800000	
1	1.4	46206.0	-4.013333	-29798.0	119589.306667	
2	1.6	37732.0	-3.813333	-38272.0	145943.893333	
3	2.1	43526.0	-3.313333	-32478.0	107610.440000	
4	2.3	39892.0	-3.113333	-36112.0	112428.693333	

squaredDeviation

0	17.752178
1	16.106844
2	14.541511
3	10.978178
4	9.692844

Final Regression Parameters
Numerator Sum: 2207082.8
Denominator Sum: 233.55466666666666

Slope (m): 9449.962321455076
Intercept (b): 24848.2039665232
Equation: y = 9449.96x + 24848.20

```
In [60]: def predictSalary(yearsInput):
        return slopeValue * yearsInput + yIntercept

# Example usage
predictedValue = predictSalary(4)
print(f"\nPredicted salary for 4 years experience: {predictedValue}")
```

Predicted salary for 4 years experience: 62648.0532523435